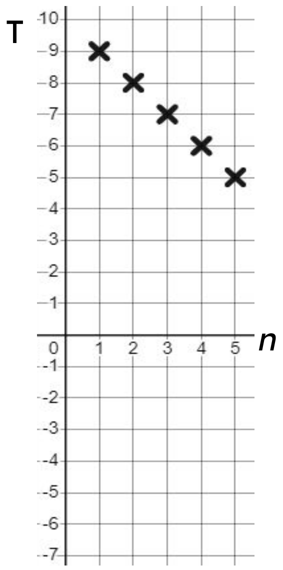
Checking and securing understanding of plotting coordinates generated from a rule

Task A: Plotting arithmetic sequences

1) Find and plot the first five terms of these arithmetic sequences.

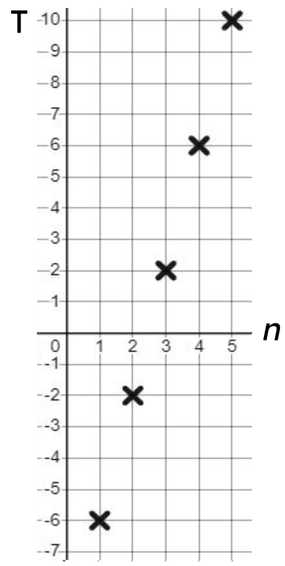
a) $4n - 10$

Term number (n)	1	2	3	4	5
Term value (T)	−6	−2	2	6	10

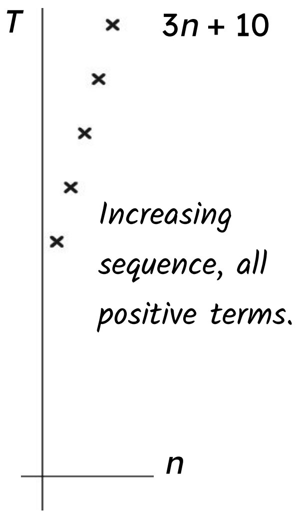


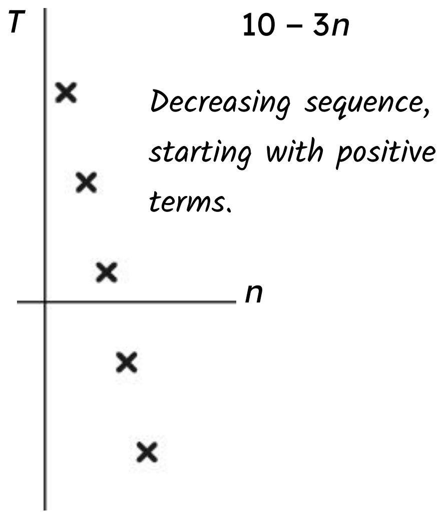
b) $10 - n$

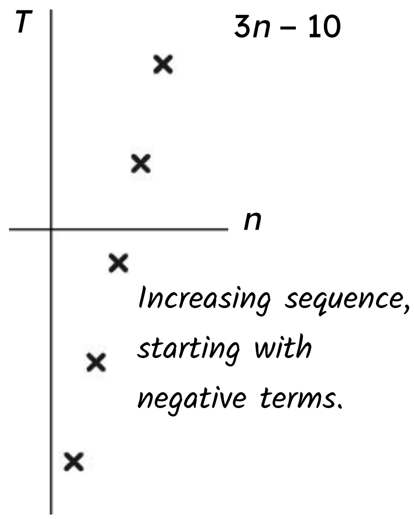
Term number (n)	1	2	3	4	5
Term value (T)	9	8	7	6	5



2) Match the sequences to their graph.







Task B: Plotting coordinates that follow a rule

1) Match the verbal rule to the simplified algebraic rule

$y = x + 3$	The x-coordinate multiplied by 5 subtracted from 3
$y = 3x + 5$	Add 3 to the x-coordinate
$y = 5x - 3$	The x-coordinate multiplied by 5 add 3
$y = 5x + 3$	The x-coordinate subtracted from 3
$y = 3 - 5x$	The x-coordinate multiplied by 5 subtract 3
$y = 3 - x$	The x-coordinate multiplied by 3 add 5

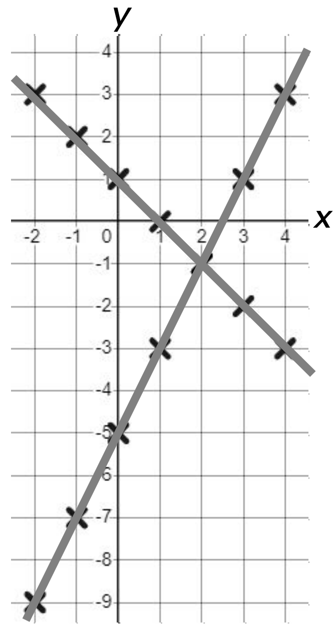
2) Complete the table of values and plot the rules:

a) $y = 2x - 5$

x	−2	−1	0	1	2	3	4
y	−9	−7	−5	−3	−1	1	3

b) $y = 1 - x$

x	−2	−1	0	1	2	3	4
y	3	2	1	0	−1	−2	−3



3) Jacob plots the rule $y = 5 - 3x$

Mark his work and give him feedback.

x	−2	−1	0	1	2	3
y	11	8	5	2	1	−4

$y = 5 - 3x$ is a linear rule, this should make a straight line.

This coordinate (2, 1) does not fit the rule. $5 - 3(2) = -1$, plot (2, −1) instead.

Also, continue your lines to the edge of your grid.